

1. Purpose

This document records what was changed in the RTL, testbench, and toolchain before generating the experimental results in Deliverable 5, and why. Per the roadmap, this step exists to decide modifications *before* running experiments — in this project's case, that decision was the single-module substitution defined in Deliverable 2.5, executed and verified across Phases 1–4 of implementation.

2. RTL Changes

File	Status	Description
<code>fixed_threshold_predictor.v</code>	New	Baseline decision logic per Deliverable 2.5: single fixed idle-cycle threshold per mode, no <code>activity_rate</code> input, no hysteresis band, memoryless w.r.t. previous state.
<code>static_pg_top.v</code>	New	Baseline top-level, structurally identical to <code>adaptive_pg_top.v</code> with <code>hysteresis_predictor.v</code> swapped for <code>fixed_threshold_predictor.v</code> . Same UPF, same ALU, same isolation/retention strategy — satisfies the Fair-Comparison Protocol (Deliverable 1).
<code>alu_32bit.v</code> , <code>activity_monitor.v</code> , <code>hysteresis_predictor.v</code> , <code>power_state_fsm.v</code> , <code>sleep_controller.v</code> , <code>isolation_ctrl.v</code> , <code>adaptive_pg_top.v</code>	Unchanged	No modifications — these remain exactly as audited in Deliverable 2 (Current System Audit).

No other RTL changes were made. The experiment isolates one variable: dual-condition adaptive hysteresis vs. fixed-threshold decision logic.

3. Testbench Changes

File	Change	Reason
<code>tb_comparison.v</code>	Rewritten to instantiate <code>static_pg_top</code> (not the original <code>static_pg_baseline</code> stub) with symmetric per-mode counters (ON/LS/DS cycles, transitions, wake-up events, max latency) tracked identically for both DUTs.	The original draft's baseline had no sleep states at all and could not lose fairly — counters were asymmetric, hardcoding the baseline to 0% savings by construction.
<code>tb_comparison.v</code> , Workload 1 (BURST)	Idle gap changed from 1 cycle to 5 cycles; burst size changed from 10 ops to 8 ops.	The original 1-cycle gap was below even the baseline's 3-cycle threshold, so neither design had any opportunity to sleep — the workload tested nothing. The corrected gap (5 cycles, exceeding the baseline's fixed threshold of 3, while keeping burst duty cycle above the adaptive design's 50% rate-widening trigger) is what actually exercises the thrashing-vs-stability distinction the workload is named for.

Across all three workloads, `tb_comparison.v`'s correctness checker (comparing `base_result` against `prop_result` whenever both DUTs report `valid` simultaneously) logged zero mismatches. This confirms the predictor substitution introduced no functional regression in the shared ALU datapath — the leakage and throughput differences reported in Deliverable 5 are attributable to decision logic alone, not to any divergence in computed results.

4. Toolchain Fixes

These are environment/script corrections discovered and resolved during implementation, not design changes, but documented here since they affect reproducibility:

Script	Issue	Fix
<code>dc_baseline.tcl</code>	Originally analyzed only <code>alu_32bit.v</code> + the old stub baseline.	Rewritten to analyze the full single-module-substitution baseline (<code>static_pg_top.v</code> + supporting modules).
<code>dc_synthesis.tcl</code>	<code>insert_clock_gates</code> (wrong command name), then <code>insert_clock_gating [current_design]</code> (wrong argument — command takes no positional design argument).	Corrected to <code>insert_clock_gating</code> with no arguments.
<code>icc2_place_route.tcl</code>	<code>read_verilog -library WORK</code> — invalid argument, not part of ICC2's syntax.	Removed; ICC2's library context comes from <code>create_lib</code> .
<code>icc2_place_route.tcl</code>	<code>create_pg_ring_pattern</code> — incorrect argument structure for this ICC2 version.	Replaced with <code>connect_pg_net -automatic</code> , sufficient for this project's scope.
<code>icc2_place_route.tcl</code>	<code>report_area</code> — disabled in this ICC2 license tier (CMD-080).	Replaced with <code>report_utilization</code> .
<code>icc2_place_route.tcl</code>	<code>write_sdc</code> with a positional filename — ICC2 requires an explicit <code>-output</code> flag (unlike DC's <code>write_sdc</code>).	Corrected to <code>write_sdc -output <path></code> .

icc2_place_route.tcl	place_opt/clock_opt failed with Scenario Manager creation failed (OPT-041) — root cause: no RC parasitic data had been attached to the design.	Diagnosed as set_tlu_plus_files (an ICC1/DC-only command, invalid in ICC2) being entirely absent rather than misused. Resolved via read_parasitic_tech + set_parasitic_parameters, pointed at the real SAED14LVT TLUplus files located at /usr/synopsys/saed/tech/star_rc/ — confirmed present and world-readable, simply never attached by any prior script.
icc2_place_route.tcl	report_power at the placement and post-route stages intermittently failed ("Errors detected during redirect"), traced to several library cells (SAEDLVT14_FSDPQM4_*, SRRDPQ4_*, etc.) that ICC2 could not infer sequential functionality for (SQM-1061), breaking its internal power model.	Wrapped both report_power calls in catch { ... } so a power-reporting failure doesn't abort placement/routing. DC's report_power (already complete and reliable) is used as the authoritative power data source instead.

5. Deviation from the Locked Research Plan

The Research Plan (Deliverable 3) specified 15 ns (low-power) / 8 ns (high-speed) design points. Actual synthesis was run at **5 ns / 2 ns**, inherited from the working scripts found during implementation and kept rather than re-run, since (a) both corners independently verified clean timing closure (MET at synthesis; WNS=0, TNS=0 post-route) for both designs, and (b) re-synthesizing both designs from scratch at different corners would have cost real toolchain time for no analytical benefit once the parasitic-data fix made the existing corners fully usable end-to-end. This is documented here as the authoritative correction to the corner values referenced in Deliverable 3; the experimental design matrix and hypothesis structure are otherwise unchanged.

6. Final Artifact Inventory (input to Deliverable 5)

- DC synthesis reports: [reports/baseline/](#), [reports/proposed/](#) — area, power, timing at both corners, both designs.
- ICC2 post-route reports: [reports/icc2/](#), [reports/icc2_baseline/](#) — real placed-and-routed area, timing, utilization, both designs, low-power corner.
- Functional comparison: [sim_out/vcs_sim_comparison.log](#) — three-workload head-to-head, real per-mode cycle counts, transitions, throughput.
- Post-route netlists and SDC for both designs ([outputs/](#)).

7. Conclusion

The RTL improvement implemented was exactly the substitution specified in Deliverable 2.5 — no scope expansion, no additional novel logic introduced mid-experiment. All toolchain issues encountered were environment/syntax corrections, documented above for reproducibility, not design changes. The project is now ready for Deliverable 5 to report results against the four hypotheses using this dataset.